Unit: 6 Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ____________
Date: __________

Homework (Habanero)

Q1. A quadratic function models the growth of a cell culture. The function is given in vertex form as $f(x) = a(x-h)^2+k$. If the vertex is at $(2, 3)$, what is the value of h and k?
(A) $h = 1, k = 2$
(B) $h = 2, k = 3$
(C) $h = 3, k = 2$
(D) $h = 2, k = 1$
(E) $h = 1, k = 3$

Q2. The quadratic function $f(x) = x^2 - 4x - 5$ can be written in intercept form as $f(x) = a(x-p)(x-q)$. What are the values of p and q?
(A) $p = -1, q = 5$
(B) $p = 1, q = -5$
(C) $p = 5, q = -1$
(D) $p = -5, q = 1$
(E) $p = 2, q = -3$

Q3. A hospital is testing the effectiveness of a new medicine. The quadratic function $f(x) = 2(x-1)^2 + 3$ models the dosage required. What is the minimum dosage required?
(A) 2 units
(B) 3 units
(C) 1 unit
(D) 4 units
(E) 5 units

Q4. The function $f(x) = x^2 - 2x - 6$ can be written in vertex form as $f(x) = a(x-h)^2 + k$. What is the value of a?
(A) 1
(B) 2
(C) -1
(D) -2
(E) 3

Q5. A quadratic function $f(x) = ax^2+bx+c$ models the statistics of patient recovery. If $a = 1$ and $b = -3$, what is the value of c if the function is written in intercept form as $f(x) = (x-1)(x-2)$?
(A) -2
(B) 2
(C) -1
(D) 1
(E) 3

Q6. The quadratic function $f(x) = 2x^2 - 4x - 3$ can be written in vertex form as $f(x) = a(x-h)^2 + k$. What is the value of h?
(A) 1
(B) 2
(C) -1
(D) -2
(E) 3

Q7. A healthcare provider is analyzing patient data using the quadratic function $f(x) = x^2 + 2x - 6$. What are the x-intercepts of the function?
(A) $(-3, 0)$ and $(2, 0)$
(B) $(-2, 0)$ and $(3, 0)$
(C) $(-1, 0)$ and $(6, 0)$
(D) $(-6, 0)$ and $(1, 0)$
(E) $(-2, 0)$ and $(-3, 0)$

Q8. The function $f(x) = 3(x-2)^2 + 1$ models the growth of a cell culture. What is the value of the function at $x = 3$?
(A) 3
(B) 4
(C) 5
(D) 7
(E) 10

Open Ended Questions (FRQ)

Q9. Write the quadratic function $f(x) = x^2 + 5x + 6$ in vertex form and intercept form. Show all work.

Q10. A quadratic function models the dosage required for a new medicine. The function is given in intercept form as $f(x) = (x-2)(x-3)$. Write the function in vertex form and find the minimum dosage required. Show all work.

Chef's Tips

- Allow the use of a calculator for complex calculations.
- Emphasize the importance of showing work for free response questions.
- Provide feedback on the interpretation of parameters in each form.